

Programming Techniques –Using C Language

Objective: The purpose of this lab is to become familiar with LOOPS

The **For loop**, **While loop** and **Do While loop**

1. Write a C program to display the message “Hello World” ten times by using a While Loop.
2. Write a C program to print all natural numbers from 1 to n using while loop.

3. Ask the user which times table they wish to see, then display it in the following format:



```
Enter an interger : 7
Times 7 table
1 * 7 = 7
2 * 7 = 14
.
.
12 * 7 = 84
Press any key to continue..
```

4. Print out all integers from 5 down to -5.
5. Write a program that prints even numbers starting from 2 to 20, for example 2,4,6,8,10,12.....,20
6. Print out the squares of the first 10 integers.
7. Write a c program to find the factorial value of a given input number.
8. Write a C program to input a number from user and count number of digits in the given integer using while loop.
9. Translate the following (Figure 01) while loop into an equivalent **for** loop.
10. Write a c program to count the no of spaces in a given line of text.
11. Modify above program to count the no of characters in a given line of text.

```
#include <stdio.h>
int main(){
    int i = 1, sum = 0, max ;
    printf("Please input number :");
    scanf("%d",&max);
    while (i <= max) {
        sum+=i ;
        i++ ;
    }
    printf ("The sum is =%d ",sum);
    return 0;
}
```

Figure 01

12. Write C programs to get the outputs as in given figures.

```
*
* *
* * *
* * * *
* * * * *
```

Figure 02

```
#####
#####
#####
####
###
##
#
```

Figure 03

13. Write programs to print the following outputs(Hint: use nested for loops)

```
1
22
333
4444
55555
```

```
1
12
123
1234
12345
```

14. Write a program to compute the xth power of y, where x is a positive integer and y a floating point number. Do not use the built in function; use a loop instead.

15. Compile, execute and check the output of following C program.

16. Write a simple loop structure to repeatedly print a message, until the user enters 'n' to stop the repetition. A sample dialogue is given in Figure 04.

1. Hello, nice to meet you.
Do you want to continue? y
2. Hello, nice to meet you.
Do you want to continue? y
3. Hello, nice to meet you.
Do you want to continue? n

Figure 04

```
#include <stdio.h>
int main(){
    int number = 44;
    int guess;
    printf("Guess a number between 1 and 100\n");
    do {
        printf("Enter your guess: ");
        scanf("%d",&guess);
        if (guess > number) {
            printf("Too high\n");
        }
        if (guess < number) {
            printf("Too low\n");
        }
    } while (guess != number);
    printf("You win. The answer is %d",number);
    return 0;
}
```

17. Write a menu-driven application to ask for two floating-point numbers, and perform addition, subtraction, multiplication and division on these two values. The user may choose which operation he would like by selecting an option on the menu. A sample run is shown in Figure 05.

```
Calculator Menu
1. Addition
2. Substraction
3. Multiplication
4. Division
5. Exit

Enter Your Choice: 1

Enter First Value : 10

Enter second value : 20

Answer : 30

Calculator Menu
1. Addition
2. Substraction
3. Multiplication
4. Division
5. Exit

Enter Your Choice: 5
```

Figure 05

18. Read a positive integer value, and compute the following sequence:
- If the number is even, halve it;
 - If it's odd, multiply by 3 and add 1.
 - Repeat this process until the value is 1, printing out each value. Finally print out how many of these operations you performed.
 - Typical output might be:

```
Initial value is 9
Next value is 28
Next value is 14
Next value is 7
Next value is 22
Next value is 11
Next value is 34
Next value is 17
Next value is 52
Next value is 26
Next value is 13
Next value is 40
Next value is 20
Next value is 10
Next value is 5
Next value is 16
Next value is 8
Next value is 4
Next value is 2
Next value is 1
Final value 1, number of steps 19
```